

REPORT STRUCTURE

Think of your LaTeX files as different “modules” in a system: each has a clear responsibility, and you route specific pieces of your RESULTS into the right place.

Below is a **section-by-section design**, including **what story to tell** and **which figures/tables to place where**.

(When I say “table” for something that’s currently just text in `RESULTS.pdf`, you can typeset it as a LaTeX table.)

0. Title / Authors / Keywords

Goal: make it clear this is a PCA + ML quality-classification study on Kepler KOIs.

- Title: mention *Kepler Objects of Interest*, *quality*, and *PCA + machine learning*.
- Keywords: “Kepler Objects of Interest, PCA, Dimensionality Reduction, Classification, Neural Networks, Random Forest, Quality Assessment”.

No figures here.

1. `sections/abstract.tex`

Goal: 1 paragraph (150–200 words) summarizing the entire pipeline and key numbers.

Content to include:

- Problem: binary quality classification of KOIs, derived from the continuous `koi_score`.
- Data: total 7,994 KOIs; after thresholding you keep 7,211 labeled examples with roughly balanced classes (3753 low / 3458 high).
- Methods: log/linear feature transforms + PCA (retain 4 PCs) + comparison of several classifiers (LR, RF, MLP, gradient boosting, etc.).
- Main PCA result: first 3 PCs explain ~73% of variance, 4 PCs >80%.

- Main ML result: best model is **MLP on original 12 features**, with test Accuracy ≈ 0.893 and F1 ≈ 0.890 .
- One sentence of interpretation: PCA gives compact structure but loses some predictive power vs using original features; a nonlinear model captures complex relations between astrophysical features and quality.

No figures or tables in abstract.

2. sections/introduction.tex

Goal: Set up *why the problem matters* and *what you contribute*; don't dive into numbers yet.

Structure:

1. Context (1–2 paragraphs)

- Brief intro to *Kepler* and *Kepler Objects of Interest* (KOIs).
- Explain that KOIs are candidate exoplanets; their `koi_score` reflects confidence/quality of the candidate vetting.

2. Problem Statement (1 paragraph)

- Frame this as a **statistical quality assessment**: given a set of astrophysical and observational features, can we automatically classify KOIs into low vs high quality labels derived from `koi_score`?

3. Challenges (short paragraph)

- Many correlated features (see later correlation matrix).
- High dimensionality; some features span several orders of magnitude (motivates log transforms and PCA).

4. Contributions (bullet list)

Something like:

- Construct a clean binary quality label from `koi_score` and perform detailed exploratory analysis.
- Apply PCA to 12 transformed features and show that a small set of PCs captures most variance while revealing underlying physical axes.
- Benchmark a range of classifiers on original and PCA-reduced features.

- Provide an in-depth analysis of the best-performing model (MLP) and feature importance / decision boundaries.

No figures here; just refer forward ("as shown in Section X").

3. `sections/data_description.tex`

Goal: Document the dataset and EDA. This is where most of the **early figures** go.

3.1 Dataset overview

- Describe:
 - Total number of KOIs, which columns are used as **features** (list the 12) and which is the **source label** (`koi_score`).
 - Mention any removed rows (missing values, invalid scores) qualitatively.
- Explain label construction:
 - Describe in words how continuous `koi_score` is converted into binary "low vs high quality" (thresholding and discarding ambiguous middle region if that's what you did).
 - Quote the final label counts (3753 low, 3458 high).

Figure 1 – KOI score distribution

- `Distribution_of_KOI_score.png`
- Caption idea: *"Histogram of koi_score with labeling thresholds indicated; low vs high quality samples are obtained by thresholding this distribution."*

Include the **summary statistics table** (count, mean, std, min, quartiles, max) as **Table 1** right next to this figure.

3.2 Feature transformations

- Explain the motivation for transforming features that span orders of magnitude or are highly skewed.
- Summarize the transformation rules:
 - $\log_{10}$: `koi_period` , `koi_duration` , `koi_prad` , `koi_teq` , `koi_insol` , `koi_srad`
 - $\log_{10}(1+x)$: `koi_depth` , `koi_model_snr`

- no transform: `koi_impact` , `koi_steff` , `koi_slogg` , `koi_kepmag`

Figure 2 – Transformed feature distributions

- `Distribution_of_Transformed_Features.png`
- Caption: *“Boxplots of transformed features after applying log/linear transformations; outliers are trimmed visually for clarity.”*

3.3 Correlation structure

- Explain in text that the features show moderate-to-strong correlations (e.g. between planetary radius, depth, and duration), motivating dimensionality reduction.

Figure 3 – Correlation matrix

- `Correlation_Matrix_of_Transformed_Features.png`
- Caption: *“Correlation matrix of the 12 transformed features, showing several groups of strongly correlated variables.”*

Optional:

Figure 4 – Pairplot (appendix or main)

- `Pairplot_of_Transformed_Features.png` if you keep it.
- Because it's visually heavy, you can either put it at the end of Data Description or in an appendix; in the main text just mention that it shows approximate separability (or lack thereof) in raw feature space.

4. `sections/methodology.tex`

Goal: Explain **how** you model the problem in a general way (math & algorithms), not yet specific numerical results.

Suggested sub-sections:

4.1 Problem formulation

- Define input vector $\mathbf{x}$ (12 transformed features), label $\mathbf{y} \in \{0,1\}$.
- Briefly state that you treat this as a supervised binary classification problem.

4.2 Principal Component Analysis

- Give a compact explanation of PCA:
 - Standardize features, compute covariance matrix, eigenvalues/eigenvectors, PCs as linear combinations.
 - Mention that you consider the first few PCs as new features, and you choose 4 PCs based on cumulative variance $> 80\%$ (you'll show the numbers later in Results).
- Optionally include the **PCA loading matrix notation** (A) but keep the big numeric matrix for the Results section.

No figures yet; keep all PCA plots for Results.

4.3 Classification models

Describe each family conceptually:

- Logistic Regression (linear decision boundary, interpretable coefficients).
- Random Forest (ensemble of trees, handles nonlinear interactions / variable importance).
- MLP Classifier (feed-forward neural network; can model complex boundaries; mention basic architecture if you know it).
- Briefly note that other models (LightGBM, etc.) are benchmarked but not analysed in depth.

4.4 Evaluation metrics

Define Accuracy, Precision, Recall, F1, AUC, Kappa, MCC; describe why F1 and AUC are important for imbalanced or cost-sensitive classification.

No figures here; equations only.

5. `sections/experimental_setup.tex`

Goal: Specify **how** you applied those methods on this dataset.

Describe:

1. Preprocessing pipeline

- Train/test split (e.g., 80/20) – mention it clearly.
- Standardization fitted on training data only, applied to test data.

- How PCA is fit: on training data transformed features, then applied to validation/test.

2. Cross-validation

- Number of folds (e.g., 10-fold) and random seeds.
- For each model, performance reported as mean $\pm$ std across folds (you later show these as tables).

3. Hyperparameter tuning

- Briefly state that you tuned LR, RF, MLP on CV (grid search / PyCaret's tuning); mention which hyperparameters at a high level (e.g. RF: number of trees, depth; MLP: hidden layers, activation).

4. Model selection protocol

- Explain that:
 - You first compare many models on the training set via cross-validation on **original 12-D features** and **4-PC features**.
 - Then you pick **three representative models** (LR, RF, MLP) for deeper analysis and evaluation on the held-out test set.

No figures needed; maybe a simple pipeline diagram if you want to create one.

6. `sections/results.tex`

Goal: Present the numerical & visual evidence. This is where **almost all plots** from `RESULTS.pdf` live, organized into logical subsections.

6.1 PCA results

Start with the eigenvalues/variance table:

Table 2 – PCA eigenvalues and explained variance

- Use the numbers from “Explained variance by component”.
- Emphasize that PC1–PC3 explain ~73.5% and PC1–PC4 ~81.8% of total variance.

Figure 5 – Scree + Pareto plot

- `ParetoPlot.png` (if you have a separate ScreePlot, you can combine as subfigures).

- Caption: *“Scree and cumulative variance plot; dashed horizontal line highlights the 80% variance threshold achieved by the first 4 PCs.”*

Then move to structure in PC space:

Figure 6 – PCA score plot (PC1 vs PC2)

- `PCA_Score_Plot.png` (or similar).
- Describe how high vs low quality KOIs distribute; note any visible separation or overlap.

Figure 7 – Biplot / loadings visualization

- `PCA_Biplot.png` and/or `Feature_Loadings_Plot.png` / `PCA>Loading_Matrix.png` .
- Use text to explain which features contribute strongly to PC1, PC2, PC3 (you already have “Top contributors by component”).

If space is tight, use **one** loadings figure (either the heatmap or the scatter plot).

6.2 Model benchmarking (cross-validation)

Here you show the big PyCaret style tables.

Table 3 – Model comparison on original 12-D features

- Use the block with Accuracy, AUC, Recall, Precision, F1, Kappa, MCC, TT(Sec).
- In the text, briefly highlight the top performers (LightGBM, GBC, RF, MLP all around 0.88–0.89 accuracy).

Table 4 – Model comparison on 4-PC features

- Use the similar table for PCA features.
- Emphasize the performance drop relative to original features but still good accuracy (~0.80).

Then focus on the three chosen models:

Table 5 – CV performance of LR, RF, MLP (original vs PCA)

- Build from the “Training on Original Features” and “Training on PCA Features” blocks.
- For each model, show mean $\pm$ std of accuracy, AUC, recall, precision, F1, etc.

- Use the text to show:
 - MLP (original) has the highest CV accuracy and F1.
 - RF is close but slightly lower; PCA versions of all models lose some performance.

6.3 Test-set performance of selected models

Table 6 – Final test performance

- Use the “Test Set Performance” table (LR, RF, MLP with original vs PCA features).
- Highlight:
 - Best accuracy: MLP original (0.8926) and F1 (0.8898).
 - RF original also strong; PCA variants systematically lower but still competitive.

6.4 Confusion matrices and ROC curves

Figure 8 – Confusion matrices

- Panel (a): `Confusion_Matrix_Original.png` (3 models).
- Panel (b): `Confusion_Matrix_PCA.png`.
- Text: describe patterns of errors — e.g., which class is more often misclassified, how this differs across models/feature sets.

Figure 9 – ROC curves (original features)

- `ROC_Curve_Original.png`.

Figure 10 – ROC curves (PCA features)

- `ROC_Curve_PCA.png`.

In the text, comment on AUC values, even if they look strangely low compared with CV (you can acknowledge potential reasons and leave deeper discussion to the Discussion section).

6.5 Decision boundaries in PCA space

Figure 11 – Decision boundaries on PC1–PC2 plane

- `Decision_Boundaries.png` with LR, RF, MLP panels.

- Explain qualitatively:
 - LR: mostly linear boundary.
 - RF & MLP: more complex, nonlinear regions; discuss how these boundaries track clusters of high/low quality KOIs.

6.6 Feature importance

Figure 12 – Random Forest feature importance (original 12-D)

- `RF_Feature_Importance_Original.png` .
- Text: highlight top features (`koi_prad` , `koi_depth` , `koi_period` , `koi_duration` , `koi_insol`) and give intuitive astrophysical interpretations (e.g., larger planetary radius / deeper transit likely correlates with more robust detections).

Figure 13 – Random Forest feature importance (PCA space)

- `RF_Feature_Importance_PCA.png` .
- Use the “Top 3 most important PCs” list to explain which PCs drive predictions.

7. `sections/discussion.tex`

Goal: Interpret and connect the results; answer “so what?” in words.

Organize around themes:

1. Dimensionality reduction vs prediction performance

- Discuss how PCA reveals underlying structure (PC1 ~ stellar environment, PC2 ~ transit depth-radius-snr, PC3 ~ size/brightness mix), but using only 4 PCs leads to lower accuracy than using all 12 features.
- Interpret why: information loss, PCA being unsupervised, class separation not fully aligned with maximum variance directions.

2. Model performance trade-offs

- Compare LR, RF, MLP:
 - LR is interpretable but underfits complex structure.
 - RF and MLP significantly improve F1 and accuracy, especially on original features.

- Comment on training time (TT(Sec) from model comparison tables) if relevant.

3. Error analysis

- Use confusion matrices to argue:
 - Whether the model tends to misclassify low quality as high (risky) or high as low (conservative).
- Relate these patterns back to practical implications in KOI vetting (e.g., false positives waste follow-up, false negatives may miss good candidates).

4. AUC discrepancy / limitations

- Reflect on the difference between CV AUC (~0.94 for RF) and test ROC curves which show $AUC < 0.5$ in your plots; mention possible causes (probability extraction, label encoding, or plotting bug) and acknowledge this as a limitation that should be addressed in future work.

5. Interpretation of feature importance

- Connect the most important features to astrophysical reasoning:
 - larger radius / deeper transit = stronger signal,
 - longer durations and periods maybe correlate with cleaner transit shapes, etc.
- Discuss how this matches or challenges expectations from exoplanet detection.

No new figures; this is mostly cross-referencing the figures already shown.

8. `sections/conclusion_future_work.tex`

Goal: Short, high-signal summary + specific next steps.

8.1 Conclusion

- Re-state objective in one sentence.
- Summarize key results in 3–4 bullet points:
 - Successful construction of binary KOI quality labels from `koi_score` with balanced dataset.

- PCA shows strong dimensionality reduction potential (first 3 PCs $\approx 73.5\%$ variance) and interpretable axes.
- Nonlinear classifiers (RF, MLP, boosting) on original features achieve ~ 0.88 – 0.89 accuracy and F1, outperforming linear baselines.
- Best model (MLP) offers a practical automatic quality screener for KOIs.

8.2 Future work

List 3–5 concrete directions:

- Fix and further investigate ROC/AUC behaviour and probability calibration.
- Explore alternative label definitions (different `koi_score` thresholds or multi-class labels).
- Incorporate additional features (e.g., time-series metrics) or domain knowledge.
- Try other dimensionality reduction methods (LDA, autoencoders) and gradient boosting models tuned more carefully.
- Consider cost-sensitive learning to explicitly penalize the more dangerous error type (decide which based on scientific context).

No figures here.

9. `sections/references.tex`

- Cite:
 - PCA / ML textbooks or standard papers.
 - Papers on Kepler KOIs and `koi_score`.
 - Libraries used (scikit-learn, PyCaret etc., in a concise way).

No figures; just standard bibliography.

If you follow this architecture, you basically “wire” every element of `RESULTS.pdf` into an appropriate place in the report, with **Data Description + Results** holding the visual evidence, **Methodology + Experimental Setup** explaining how you got there, and **Discussion + Conclusion** explaining what it all means.